

Yarn AI

Pricing & Business Model — Nigerian Brand Voice Engine

Prepared 14 June 2026 · Assumptions: FX ₦1,600/\$1; everyday generations on Sonnet 4.6 / Haiku 4.5, premium polish on Opus 4.8 (model tiering). All figures are planning estimates, not guarantees.

Claude API pricing (cost basis)

Model	Input (\$/1M tok)	Output (\$/1M tok)
Opus 4.8 (claude-opus-4-8)	\$5.00	\$25.00
Sonnet 4.6 (claude-sonnet-4-6)	\$3.00	\$15.00
Haiku 4.5 (claude-haiku-4-5)	\$1.00	\$5.00

1 • Operating cost scenarios

1a. Unit cost per action

Based on real token usage in the app (system prompt ~1,200 input tokens; output varies by feature).

Action	Tokens (in / out)	Opus 4.8	Sonnet 4.6	Haiku 4.5
Generation (3 variants)	1,200 / ~1,400	₦70	₦23	₦7
Refine one option	1,000 / ~600	₦40	₦12	₦4
Content calendar (12 posts)	1,200 / ~4,500	₦210	₦90	₦30
Rate / Brand advisor	1,300 / ~4,000	₦190	₦80	₦27

Takeaway: the same generation costs ₦70 on Opus but ₦23 on Sonnet and ₦7 on Haiku — a 3–10x reduction. Defaulting everyday work to Sonnet/Haiku is what makes the plans below profitable.

1b. Cost per active user / month

Most users use 30–50% of quota, not 100%. Modeling a Growth user at ~250 generations/month:

Model used	Cost/mo @250 gens	vs ₦19,000 Growth price
All Opus	₦17,500	~8% margin (loss-making)
All Sonnet	₦5,750	~70% margin
All Haiku	₦1,875	~90% margin

1c. Three operating-cost scenarios (total monthly burn incl. AI)

Stage	Users	Monthly burn	Notes
Lean / pre-revenue	≤500 (mostly free)	₦60k–₦150k	Hosting + light AI
Base / traction	~5,000 (~480 paid)	₦3.5M–₦4M	AI-dominated; ~71% gross margin
Scale	~50,000 (~4k paid)	₦28M–₦34M	~70% margin WITH tiering; ~30% without

2 • Recommended freemium model

Free tier wide enough to prove the moat (the Naija voice) on the user's own brand, then gate the stickiest features — calendar, multiple brands, premium quality — behind paid. Anchor pricing to what it replaces: a freelance writer (₦5–20k/campaign) or a junior social media manager (₦50–150k/mo).

Tier	Price/mo	Generations	Brands	Calendar	Model	Gross margin*
Free	₦0	10	1	No	Haiku	— (acquisition)
Starter	₦7,500	150	1	No	Sonnet	~85%
Growth (hero)	₦19,000	600	3	Yes	Sonnet	~70%
Pro	₦45,000	Unlimited	10	Yes	Sonnet + Opus polish	~75%
Agency	₦150,000	Unlimited	25–50	Yes + white-label	Sonnet + Opus	~80%

*at realistic 30–50% quota utilization. “Unlimited” = fair-use cap.

Plus two non-subscription layers:

- Credit packs (“airtime model”): ₦5,000 = ~50 generations, no subscription. Captures occasional users + agency overflow; matches how Nigerians buy data/airtime. ~80% margin.
- Done-for-you tier (services): ₦80k–₦200k/mo per client — your team runs their socials on the platform. Funds early growth, generates testimonials, teaches you what converts.

Annual = 2 months free (e.g. Growth ₦190,000/yr) to lock cash and cut churn.

3 • Projected earnings (optimal plan mix)

Paid mix Starter 45% / Growth 35% / Pro 15% / Agency 5% → blended ARPPU ≈ ₦24,000/mo. Monthly churn ~6% (base).

Scenario	Month	Free actives	Conv.	Paid	MRR	ARR	Gross profit/mo
Conservative	M12	3,000	4%	~120	₦2.9M	₦35M (~\$22k)	~₦2.0M
Base (hero)	M12	8,000	6%	~480	₦11.5M	₦138M (~\$86k)	~₦9M
Optimistic	M12	20,000	8%	~1,600	₦38M	₦461M (~\$288k)	~₦28M
Base	M24	30,000	7%	~2,100	₦50M	₦600M (~\$375k)	~₦40M

Where the money concentrates (the “optimal plans”):

- Growth + Pro = ~60–65% of subscription revenue despite being a minority of users. Make Growth the hero.
- Agency is highest-leverage: 20 agencies on ₦150k = ₦3M/mo at ~80% margin, near-zero marginal cost, low churn. One signed agency ≈ 8 Growth users in revenue.
- Credits add ~10–15% on top of subscription MRR and convert occasional users into subscribers.

4 • Addressable market (TAM / SAM / SOM)

Nigeria first, then pan-African expansion. Anchors: ~39–40M MSMEs (SMEDAN/NBS), a large youth-driven creator & gig economy, and a growing digital-ads market.

Segment sizing (Nigeria)

Segment	Population	Reachable	Willingness	Annual value (reachable)
Online-active SMEs	~5M of 40M MSMEs	~1.5M	₦12k/mo	~₦216B (~\$135M)
Creators / freelancers	~3M	~600k	₦8k/mo	~₦58B (~\$36M)
Marketing agencies / studios	~5,000	~1,500	₦150k/mo	~₦2.7B (~\$1.7M)

TAM → SAM → SOM

Layer	Definition	Nigeria	USD
TAM	All online-capable SMEs + creators + agencies	~₦1.2T/yr	~\$750M
SAM	Smartphone + Pidgin/English + willing to pay (~15% of TAM)	~₦180B/yr	~\$110M
SOM	Realistically capturable in 3 years (~1–3% of SAM)	~₦2–5B/yr ARR	~\$1.5–3M

The Base M24 projection (~₦600M ARR) is ~12–30% of the 3-year SOM — aggressive but achievable, with large headroom.

Expansion multiplier: Pidgin/voice localization ports to Ghana, Kenya, South Africa, Côte d'Ivoire. Pan-African market is ~3–4x Nigeria → blended TAM ~\$2.5–3B. The moat (localized voice engine) is the asset that ports across markets while staying defensible vs. global tools.

TL;DR

- Margins work only with model tiering — Sonnet/Haiku default keeps gross margin ~70–85%; all-Opus loses money on heavy users.
- Free → Starter ₦7,500 → Growth ₦19,000 (hero) → Pro ₦45,000 → Agency ₦150,000, plus ₦5,000 credit packs.
- Base case: ~₦138M ARR by M12, ~₦600M by M24, driven by Growth/Pro + agency white-label.
- TAM ~\$750M (Nigeria) / ~\$2.5–3B (pan-African); SAM ~\$110M; 3-yr SOM ~\$1.5–3M ARR.